History of Computers

How we got here

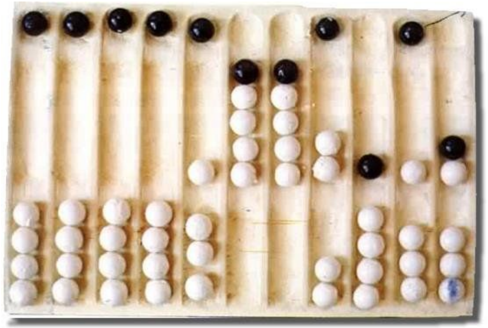
Where do we start?

- What **IS** a computer?
- Nowadays we think of something like this ——---------->
- But the word has been around longer than we've had light bulbs
    - ~1802 - first light bulb
    - ~1640's- first definition of computer I could find
- Simply put: A computer is an agent that computes
    - Now we mean a computing machine, but back then it almost universally referred to a human doing repeated mathematical operations for other disciplines

The Prehistory (~3000 BCE to ~0 CE)

- We have been counting and doing math for thousands of years
  - The Sumerians are the first civilization we have credible (~3000 BCE) evidence showing advanced mathematical understanding - We still use their number system today!

- All that time made us able to find shortcuts and ways of automating the tedious and annoying mental work
  - And since we are toolmakers by design, there is a history of the devices we used to do these tasks



The Prehistory (~100 BCE)

- The devices used to assist in mathematical operations prior to the Common Era were basic and were used to aid in counting and reference, with the human computers putting it all together to do something useful
  - Like business, astronomy, or building

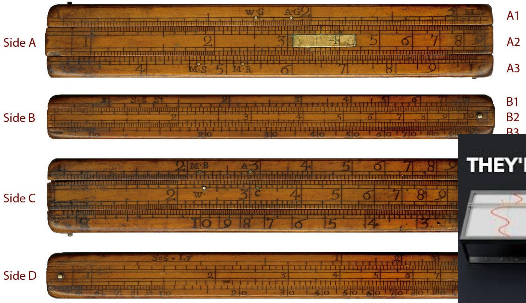
- However artifacts like the Antikythera mechanism makes historians scratch their heads at its complexity

Basic Analog Computers (0 - ~1800's)

- Analog computers refer to computing devices that utilize physical phenomena to model a problem at hand
    - Phenomena like gears, water falling, springs, and pulleys

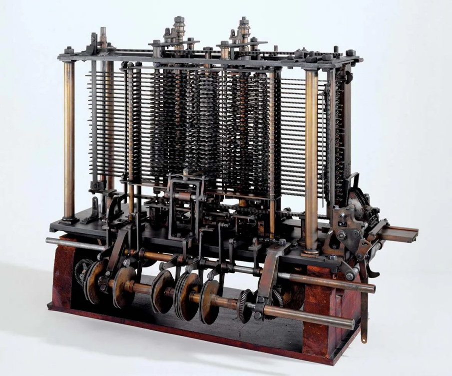
- They can only calculate singular things
    - Like known trig values, position on the earth, time, paths of planets, etc
    - Cannot change problem to something else; **not general**

- Planispheres, astrolabes, and various slide rulers are examples of simple analog computers

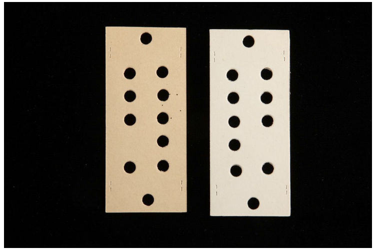


“First” Mechanical Computer (~1837) and Programmer(~1840)

- Though he never made a working prototype, Charles Babbage is attributed with creating the first general computer
  - He was making a mechanical analog computer but realized it could be more general
  - Originally the Analytical Engine was supposed to be restricted to calculating polynomial functions



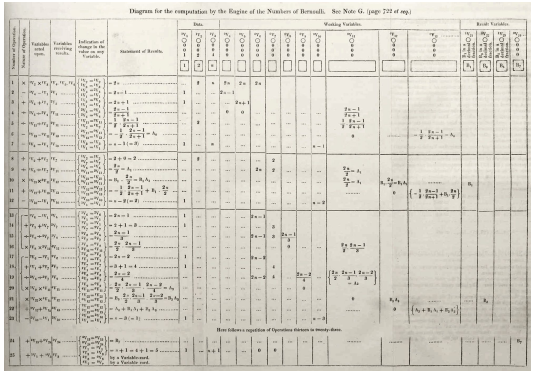
- Taking inspiration from automatic looms at the time, Babbage used punch cards to input new functionality



“First” Mechanical Computer (~1837) and Programmer(~1840)

- Because a fully realized Analytical Engine wasn’t made until much later, it existed at the time as a proposed machine that others could learn about

- Ada Lovelace read a lecture Babbage had given on it and realized the power from Babbage’s idea and created the first computer program
    - It calculated the Bernoulli numbers, something that the Analytical Engine and other analog machines could only do if the crafter made it that way



Analog VS Digital Computers

Analog Computers

- Utilizes natural, physical phenomenon to simulate problems in other domains
- Fixed programs
- No memory
- Interacting with it is not intuitive since you are working with gears, balls, and pulleys that are specifically designed for a single purpose
- Special purpose

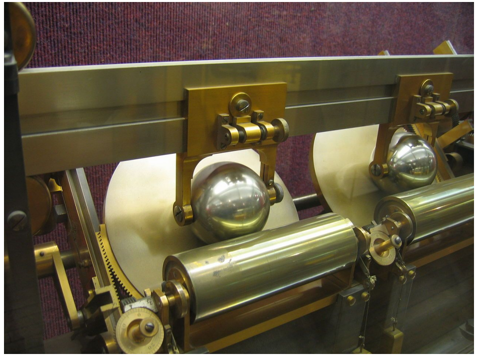
Digital Computers

- Abstracts problem solving into discrete parts (ie logical building blocks) that can be programmed to do different operations
- Stored programs
  - Unlike the tide predictors, digital computers can change their behavior without rebuilding
- Expandable memory
- Better interactivity
- General

Digital on the Rise (early 1900’s)

- Others added on to Baggage’s concept, improving the design and making prototypes to facilitate his intended use

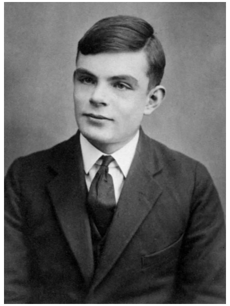
- General computer’s didn’t catch on until the early 1900’s, up until then more and more sophisticated analog devices were created
  - They were able to manage factory machines, predict weather, and assist mathematicians
  - More complex designs were developed to replace pulley and water based systems with electrical gears

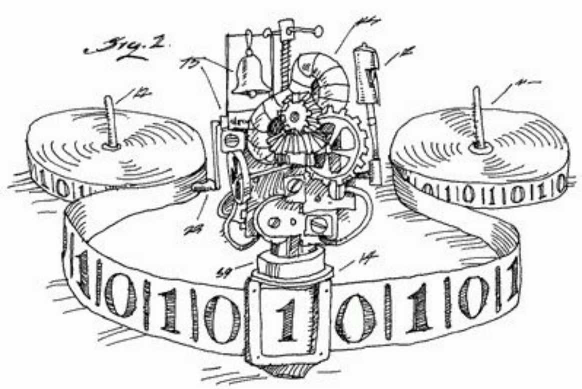


Birth of the Idea of the Modern Computer (1936)

- In 1936 Alan Turing published *On Computable Numbers,* a paper exploring what is computable in math
  - In it he proved that there are limitations to what we can know about a computing machine

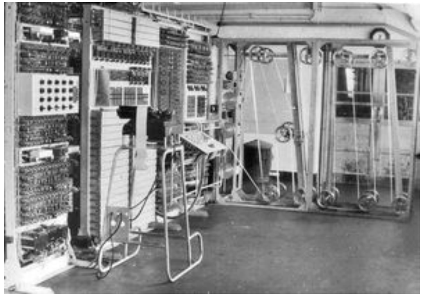
- In order to reach his conclusions he conceptualized the idea of the Turing machine





Digital Revolution (1930's)

- Digital computers spread into many domains during World War II
  - The race for technological dominance on the battlefield drove scientists to find ways to apply the powerful technology of digital computers



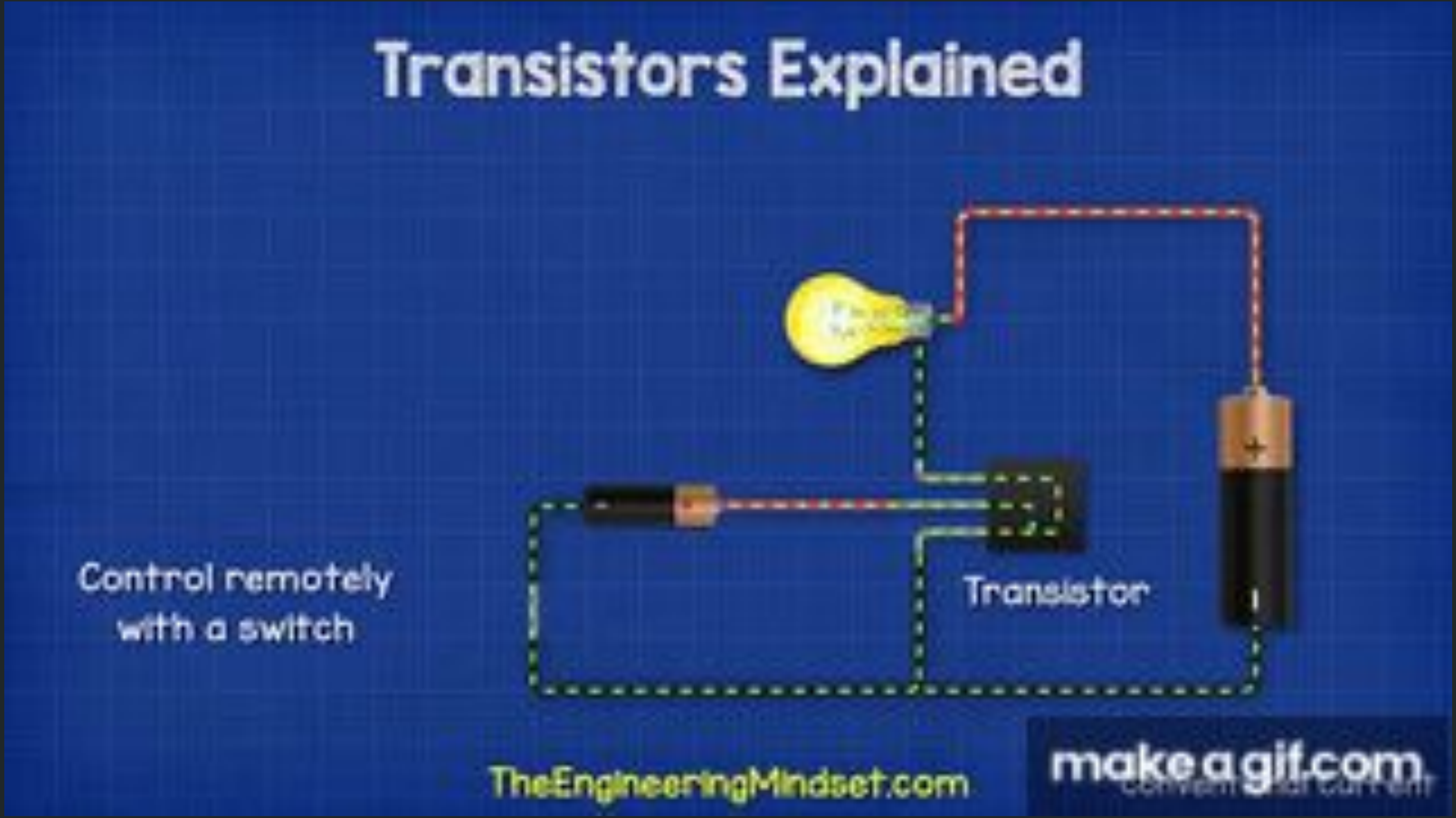
- The move from electronic gears to vacuum tubes made computers easier to store and faster

Digital Revolution (1940's)

- Bell Labs develop the transistor in 1947, a much smaller electrical component that had the same function as a vacuum tube

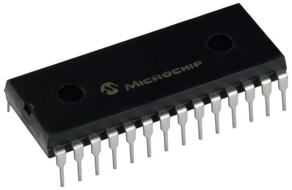
- Transistors and vacuum tubes are essentially electronic switches that either block or allow the flow of electricity

- 1949 saw the development of of Assembly Language to make programming the complex network of transistors easier

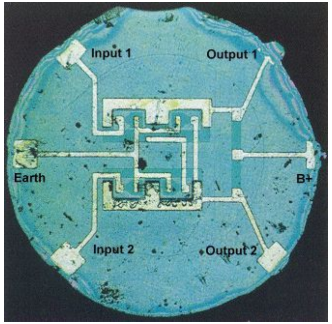

Transistors Explained
Control remotely
with a switch
Transistor
TheEngineeringMindset.com
makeagif.com

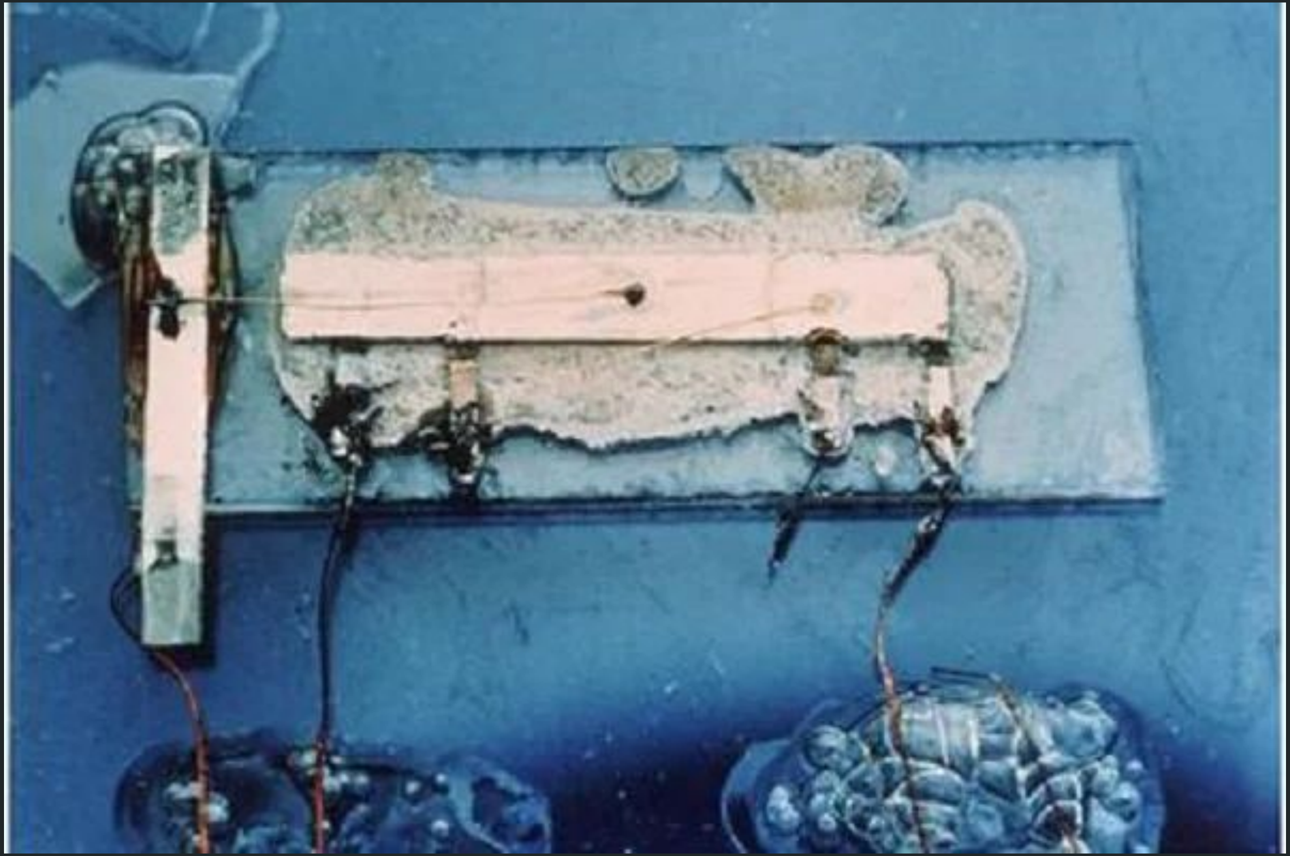
Digital Revolution (1950's onward)

- Transistors continued to improve, getting smaller at a very fast rate



- They became so small that it became infeasible to work with individual transistors leading to the development of Integrated Circuits (ICs) (1952-1959)
    - These also allowed for general or specific purpose chips to be made, giving digital logic to the electronics industry





1959 - Jack Kilby at Texas Instruments

Digital Revolution (1950's onward)

- As computers rapidly became more complex and harder to work with, researchers needed a better way to communicate with the logical operations

- The first widely used programming languages were developed as a medium between human users and the sophisticated machine and assembly code

- COBOL (1953) = **CO**mmon **B**usiness-**O**riented **L**anguage
- FORTRAN (1954) = **FOR**mula **TRAN**slation

Binary

```
0011111101100000000000000011110000
00011010010000011110011111000100001
11110011110010111110011010000001101
0001001111110001011001110011100000
000100110000000000010011110001100000
00010011001100010010011110011100000
00011010010000011110011111000100001
001101000000000100111000000011001000
000100110000000000010011110001100000
000100111111000000010001110011110000
```

Hexadecimal

```
0000000 0000 0001 0001 1010 0010 0001 0004 0128
0000010 0000 0016 0000 0028 0000 0010 0000 0020
0000020 0000 0001 0004 0000 0000 0000 0000 0000
0000030 0000 0000 0000 0010 0000 0000 0000 0204
0000040 0004 8384 0084 c7c8 00c8 4748 0048 e8e9
0000050 00e9 6a69 0069 a8a9 00a9 2828 0028 fdfc
0000060 00fc 1819 0019 9898 0098 d9d8 00d8 5857
0000070 0057 7b7a 007a bab9 00b9 3a3c 003c 8888
0000080 8888 8888 8888 8888 288e be88 8888 8888
0000090 3b83 5788 8888 8888 7667 778e 8828 8888
00000a0 d61f 7abd 8818 8888 467c 585f 8814 8188
00000b0 8b06 e8f7 88aa 8388 8b3b 88f3 88bd e988
00000c0 8a18 880c e841 c988 b328 6871 688e 958b
00000d0 a948 5862 5884 7e81 3788 1ab4 5a84 3eec
00000e0 3d86 dcb8 5cbb 8888 8888 8888 8888 8888
00000f0 8888 8888 8888 8888 8888 8888 8888 0000
0000100 0000 0000 0000 0000 0000 0000 0000 0000
*
0000130 0000 0000 0000 0000 0000 0000 0000
000013e
```

Assembly Hello World

```
    .equ    LAST_RAM_WORD,    0x007FFFFC
    .equ    JTAG_UART_BASE,   0x10001000
    .equ    DATA_OFFSET,      0
    .equ    STATUS_OFFSET,    4
    .equ    WSPACE_MASK,      0xFFFF

    .text
    .global _start
    .org    0x00000000

_start:
    movia       sp, LAST_RAM_WORD
    movi        r2, '\n'
    call        PrintChar
    movia       r2, MSG
    call        PrintString
_end:
    br          _end

PrintChar:
    subi        sp, sp, 8
    stw         r3, 4(sp)
    stw         r4, 0(sp)
    movia       r3, JTAG_UART_BASE
pc_loop:
    ldwio       r4, STATUS_OFFSET(r3)
    andhi       r4, r4, WSPACE_MASK
    beq         r4, r0, pc_loop
    stwio       r2, DATA_OFFSET(r3)
    ldw         r3, 4(sp)
    ldw         r4, 0(sp)
    addi        sp, sp, 8
    ret

PrintString:
    subi        sp, sp, 12
    stw         ra, 8(sp)
    stw         r3, 4(sp)
    stw         r2, 0(sp)
    mov         r3, r2
ps_loop:
    ldb         r2, 0(r3)
    beq         r2, r0, end_ps_loop
    call        PrintChar
    addi        r3, r3, 1
    br          ps_loop
end_ps_loop:
    ldw         ra, 8(sp)
    ldw         r3, 4(sp)
    ldw         r2, 0(sp)
    addi        sp, sp, 12
    ret

        .org        0x1000
MSG:    .asciz      "Hello World\n"
        .end
```

```c
/*
* C Program to Print "Hello World"
*/
#include <stdio.h>

int main(){
    printf("Hello World");
    return 0;
}

//Site : BTechgeeks.com
```

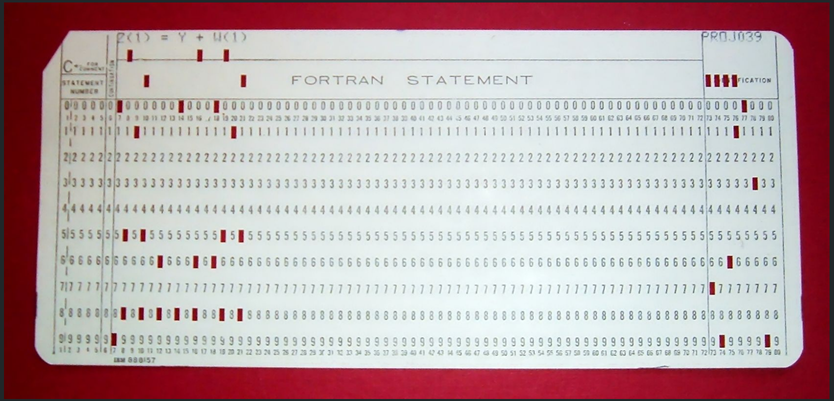
Modern Languages Hello World

```python
1  print("Hello, World!")
```

```
Hello, World!
[Finished in 0.2s]
```


helloworld.f (~) - VIM
program helloworld
implicit none
character*13 hello_string ! This is a comment.
c This is also a comment.
hello_string = "Hello, world!"
write (*,*) hello_string
end program helloworld
"helloworld.f" 7L, 221C
7,28
Alles


Command ===>
Scroll ===> CSR
=COLS> ----+----1----+----2----+----3----+----4----+----5----+----6----+-
****** ***************************** Top of Data ******************************
000100 000100 IDENTIFICATION DIVISION.
000200 000200 PROGRAM-ID. HELOWOLD.
000300 000300 ENVIRONMENT DIVISION.
000400 000400 PROCEDURE DIVISION.
000500 000500 DISPLAY 'HELLO WORLD'.
000600 000600 STOP RUN.
****** **************************** Bottom of Data ****************************




```fortran
      C
              PIN=0.02
              IF(DDT.NE.0.0)  THEN
              DT=DDT
              ELSE
              DT=PIN
              ENDIF
              WRITE(*,'(A)') '    PLEASE ENTER NAME OF OUTPUT FILE (FOR EXAMPLE
             *  B:ZZ.DAT)'
              READ(*,'(A)') FNAMEO
              OPEN(6,FILE=FNAMEO,STATUS='UNKNOWN')
              PV=WFLX/TH
              RS=NEQ*ROU*KD/TH
              CO=CS
      C
              TIME=0.0D0
              EF=0.0D0
        5     CONTINUE
              GAMMA=DT/(2.D0*DX*DX)
              BETA=DT/DX
              IF((BETA*PV).GT.0.50D0) GO TO 7
              IF((GAMMA*D/(BETA*PV)).LT.0.5D0) GO TO 6
              GO TO 8
        6     DX=DX/2
              GO TO 5
        7     DT=DT/2
              GO TO 5
        8     CONTINUE
              N=COL/DX
              NM1=N-1
              NM2=N-2
              NP1=N+1
              GAMMA=DT/(2*DX*DX)
```

```cobol
IDENTIFICATION DIVISION.
       PROGRAM-ID. ADD_NUMBERS.
       DATA DIVISION.
       FILE SECTION.
       WORKING-STORAGE SECTION.
       01  FIRST-NUMBER    PICTURE IS 99.
       01  SECOND-NUMBER   PICTURE IS 99.
       01  RESULT          PICTURE IS 9999.
       PROCEDURE DIVISION.

       MAIN-PROCEDURE.
           DISPLAY "Here is the first Number "
           MOVE 8 TO FIRST-NUMBER
           DISPLAY FIRST-NUMBER

           DISPLAY "Let's add 20 to that number."
           ADD 20 TO FIRST-NUMBER
           DISPLAY FIRST-NUMBER

           DISPLAY "Create a second variable"
           MOVE 30 TO SECOND-NUMBER
           DISPLAY SECOND-NUMBER

           *>COMMENT: COMPUTE THE TWO NUMBER AND PLACE INTO RESULT*
           COMPUTE RESULT = FIRST-NUMBER +  SECOND-NUMBER.

           DISPLAY "The result is:".
           DISPLAY RESULT.
        STOP RUN.
END PROGRAM ADD_NUMBERS.
```

Digital Revolution - Interactivity (1960’s)

- Researchers were getting tired of the interfaces they were using (ie. punch cards and binary switches)

- 1968 Douglas Engelbart gave “The Mother of All Demos”, showing off the new way to interact with
    - Basically the first demonstration of what we would recognize as a modern computer
    - Even had rudimentary voice commands

Digital Revolution - Fundamental Software (1970s)

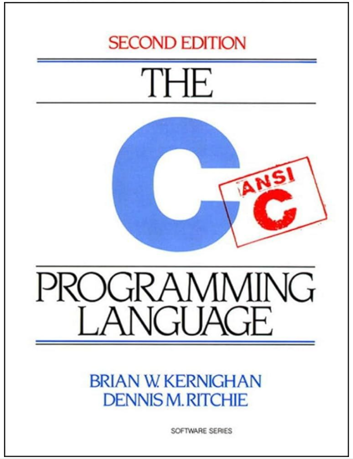
- To further make working with computers easier, the Unix Operating System (OS) began development in 1969

- To assist writing utilities for the Unix OS Bell Labs developed the C language in 1972

- Between the development of the interface in the 1960's and these advancements, the beginning of our current understanding of computers began to take form

UNIX®

An Open Group Standard

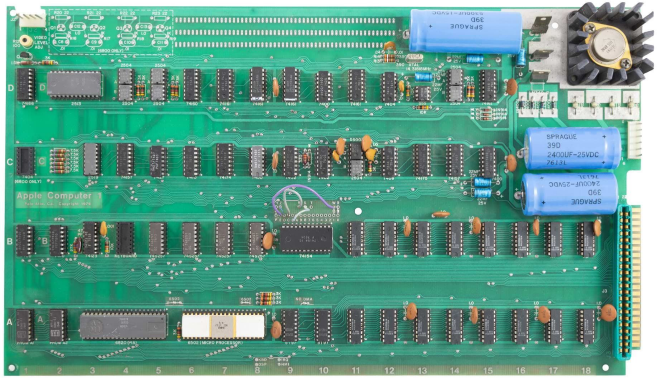


Microprocessors (1970's)

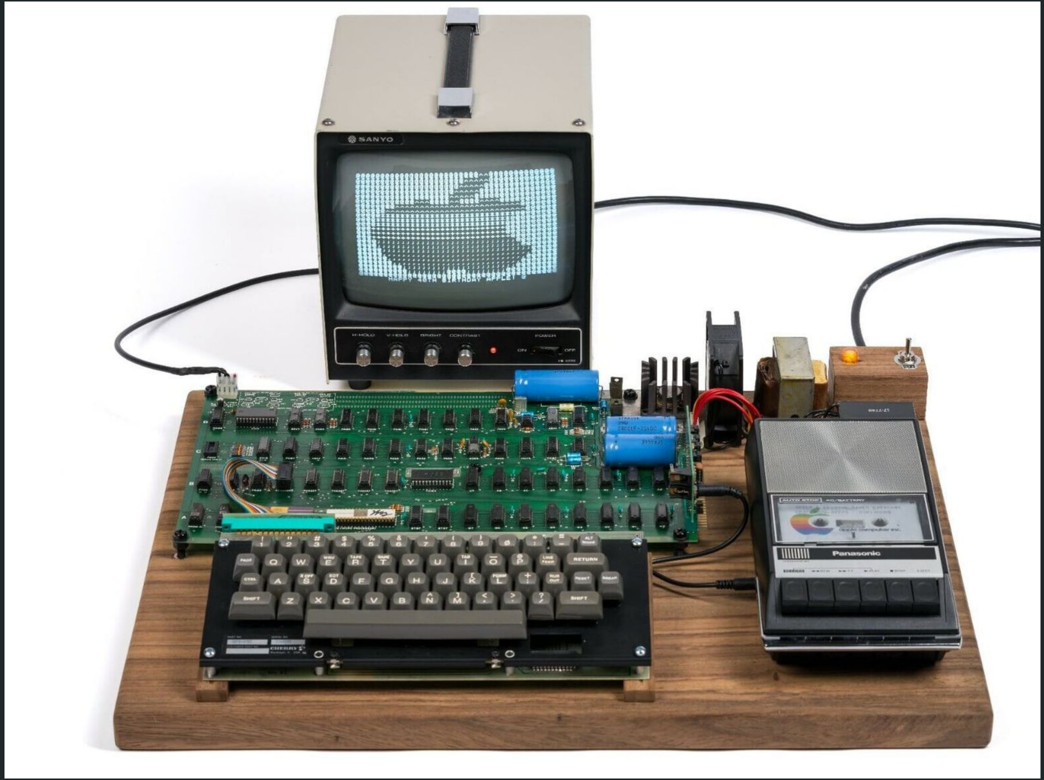
- ICs were added on to, getting more transistors and more power into the hands of computer scientists

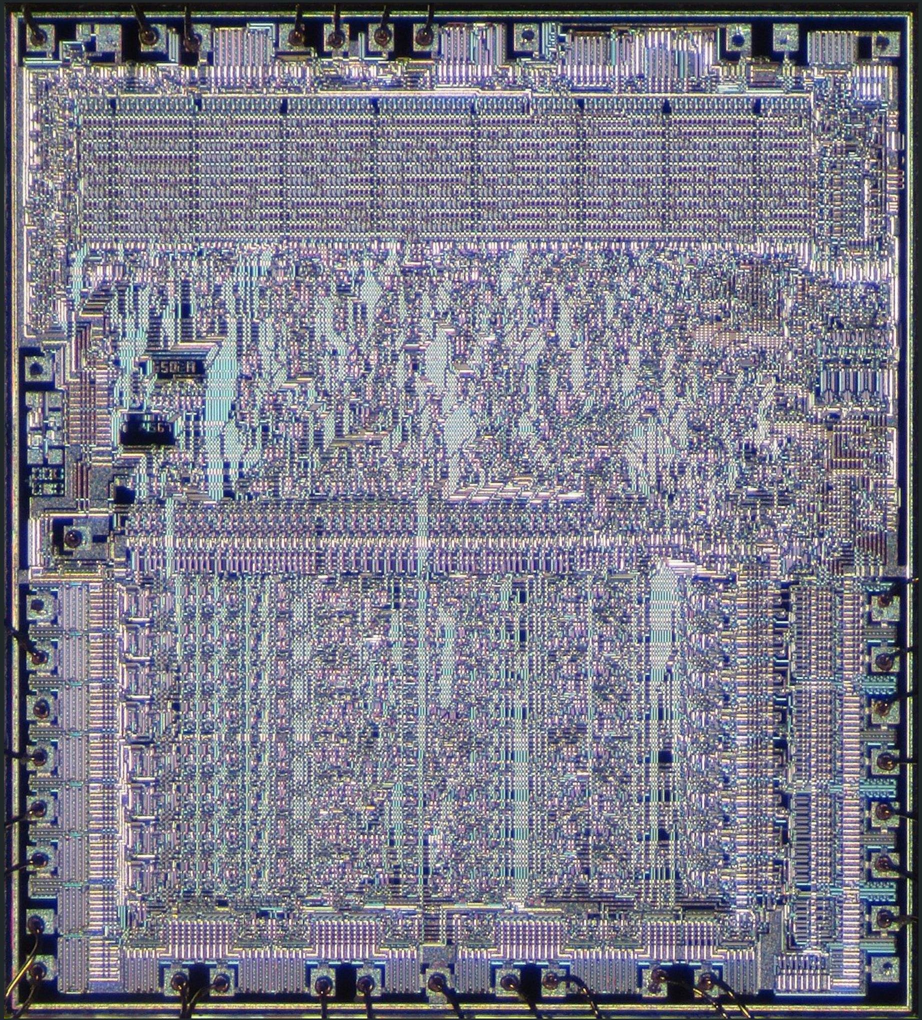
- The number of transistors grew to a point where a single IC became known as a Microprocessor
    - This paired with the better interaction medium from the previous slide, gave way to personal computers

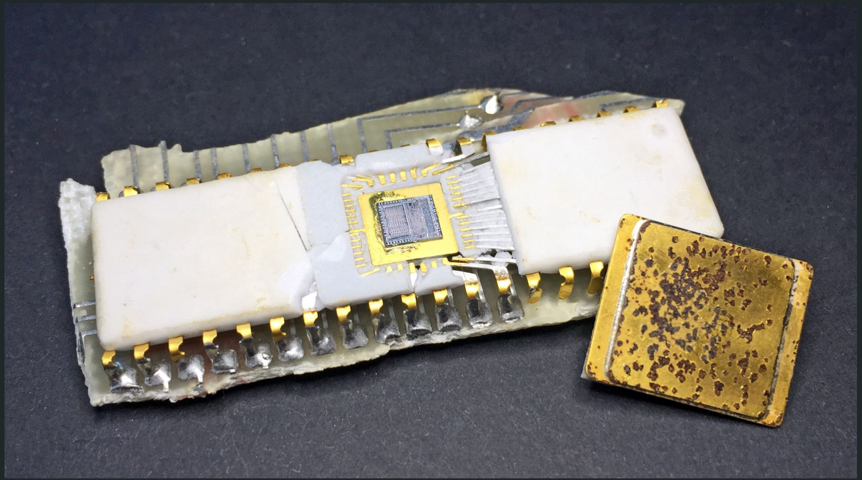
- 1976 - Steve Wozniak releases the Apple 1




APPLE
COMPUTER


SANYO
Panasonic





Moor’s Law Takes off (1980’s - 2000’s)

- In the 80’s personal computers were developed and released in rapid succession

- Businesses realized the value of them for automating tasks and offloading complex math to computer divisions
  - Start of business culture surrounding computers

- This is also the time where computers were being used for more than just math and business
  - Commodore 64 is released in 1982


1980
INTELLIVISION
Computer TV-GAME
GAME&WATCH
1981
Commodore VIC-20
IBM PC.
sinclair ZX81
CASSETTE VISION
PHILIPS
VIDEOPAC G7200
1982
ATARI Super System
ATARI Video Computer System
COLECOVISION
GEMINI
INTELLIVISION II
Vectrex
Commodore 64
ARCADIA-2001
sinclair ZX Spectrum
1983
ADAM
AQUARIUS
apple IIe
FAMILY COMPUTER
SEGA
SG-1000
ATARI JAPANESE 2600 SYSTEM
PHILIPS
VIDEOPAC+ G7400
ORIC-1
SONY MSX
1984
Apple Macintosh
SEGA
Super Cassette Vision
ORIC
AMSTRAD CPC 464
AMSTRAD CPC 664
1985
Nintendo
ENTERTAINMENT SYSTEM
ATARI ST
AMIGA
IntV System III
SEGA MARK III
sinclair
Spectrum+
AMSTRAD CPC 6128
1986
ATARI Junior.
SEGA
Master System
7800 ProSystem
DISK SYSTEM
FAMICOM
sinclair
Spectrum+2
1987
ATARI XE
AMIGA A500
sinclair
Spectrum+3
NEC PC Engine
1988
NEC PC Engine CD-ROM
MEGA DRIVE
1989
LYNX
SEGA
GENESIS
NEC
TURBOGRAFX
GAME BOY
NEC SUPER

Moor’s Law Takes off (1980’s - 2000’s)

- Microprocessors get smaller and smaller, fitting more and more transistors into their form factor

- Moore's Law basically states that, the number of transistors that we can fit within a certain area doubles about every two years
    - This is exponential growth for you math-minded folks

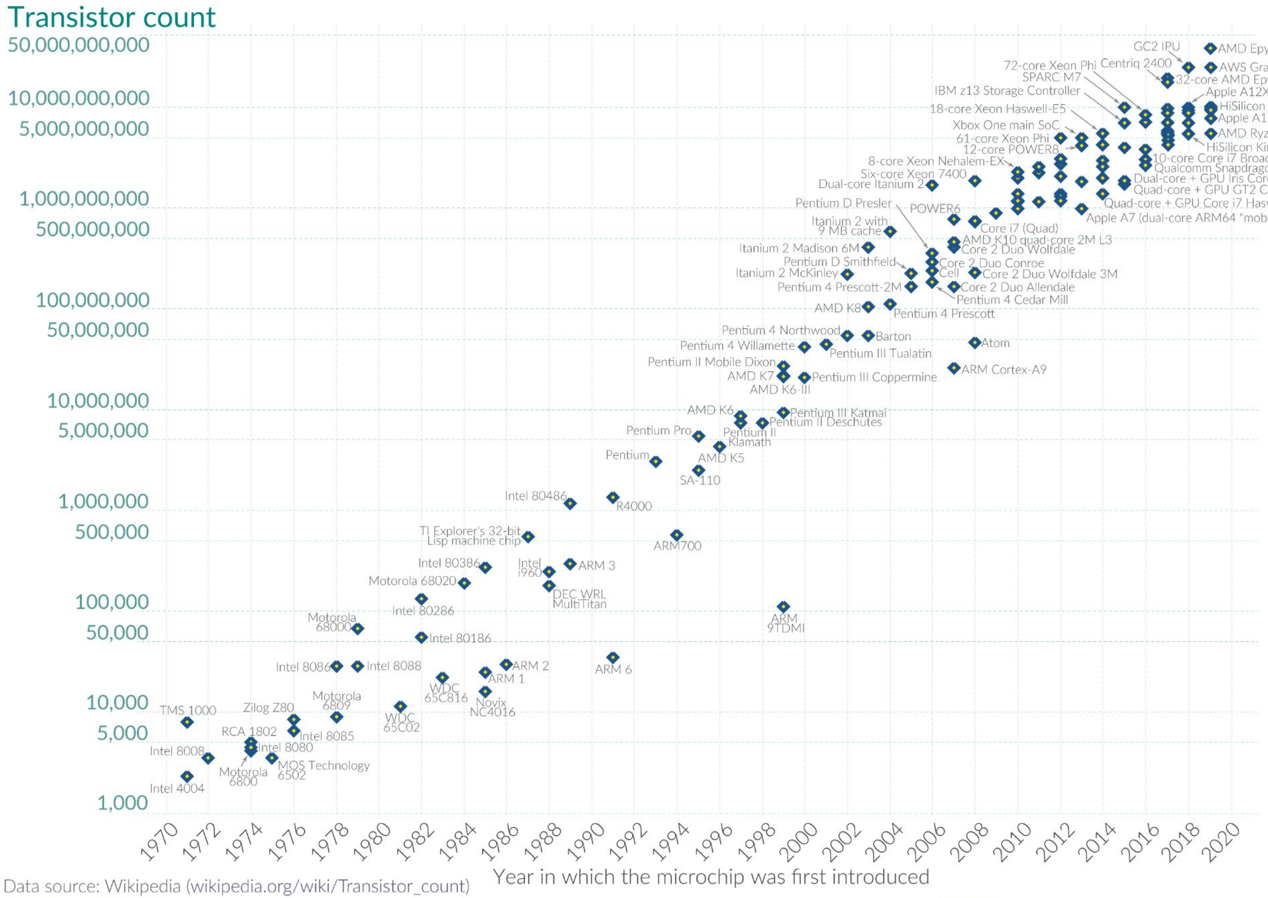
- This is an observation and not a law of nature but during this time and in recent years it is still a good predictor
- Intel releases their first Pentium CPU in 199

Wikipedia Appaloosa

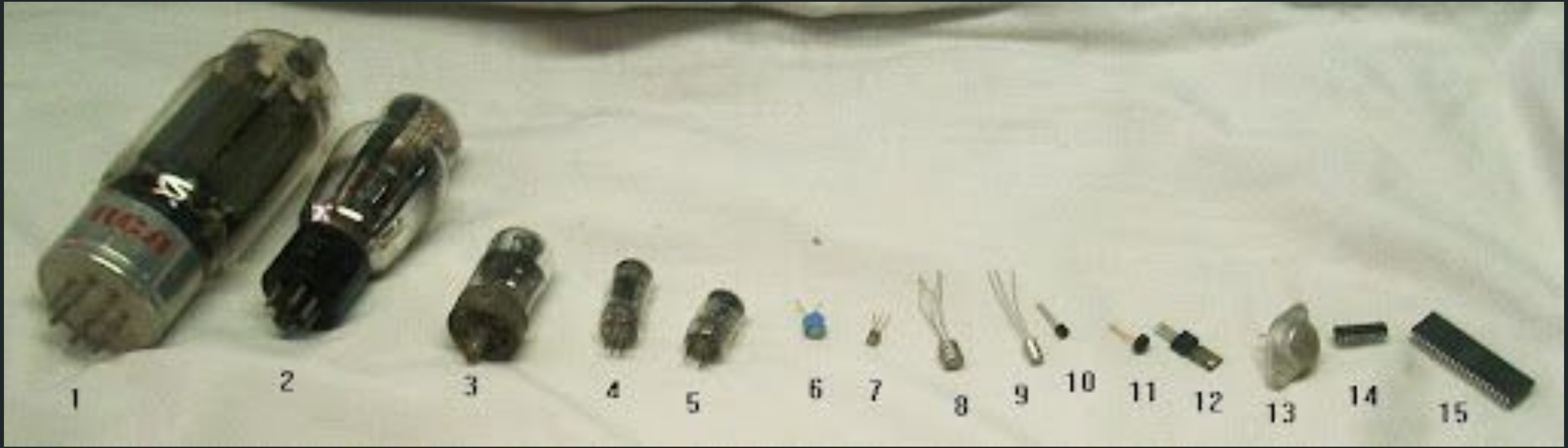
Moore's Law: The number of transistors on microchips doubles every two years

Our World in Data

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years.
This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)
OurWorldinData.org – Research and data to make progress against the world's largest problems.


1
2
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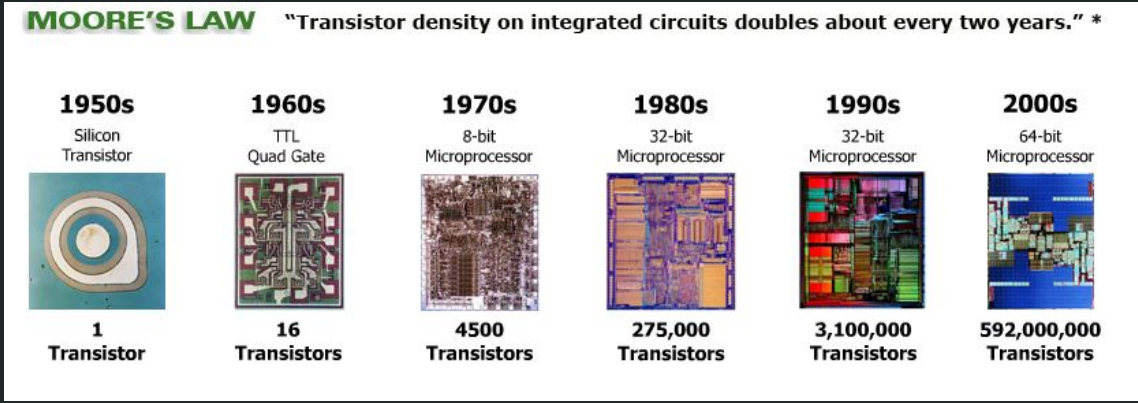
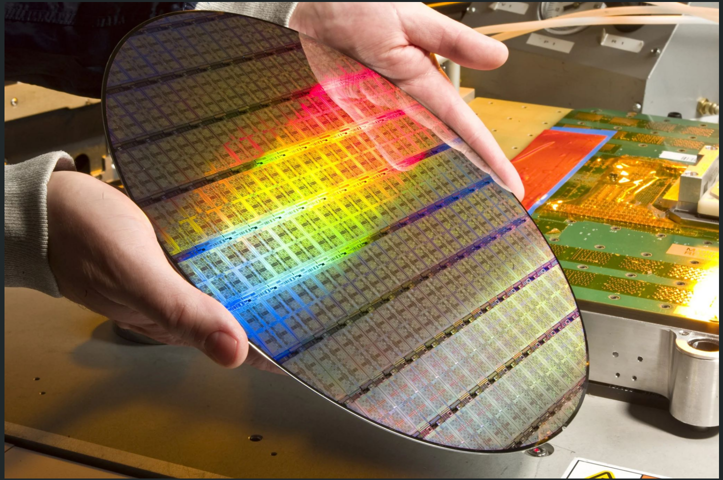

MOORE'S LAW
"Transistor density on integrated circuits doubles about every two years." *
1950s
Silicon Transistor
1 Transistor
1960s
TTL Quad Gate
16 Transistors
1970s
8-bit Microprocessor
4500 Transistors
1980s
32-bit Microprocessor
275,000 Transistors
1990s
32-bit Microprocessor
3,100,000 Transistors
2000s
64-bit Microprocessor
592,000,000 Transistors



The Internet Age (1990’s - 2000’s)

- This hasn’t been a history of the Internet or how we connect computers, but it has been in the background
- The 1990’s is when the idea of a “World Wide Web” enters the public zeitgeist
- 1993 - Mosaic, the first web browser is released
- 1999 - the abbreviation of WiFi is coined for “wireless fidelity”

We kinda know it from here (2000's - Today)

- Computers just kept getting smaller, to the point where they are in our pockets

- Everything now goes through a computer some kind
  - Banking, medical, sciences, telecommunications, national parks, and our government all run on computers

- The newest revolution that we have been seeing is the AI Revolution, due to the focus on improving GPUs rather than CPUs